

Phase 2: Project Structure Generation

1. System Analysis

System Type: Web app

Features:

- User Registration and Login
- Restaurant Browsing
- Order Placement
- Real-time Order Tracking
- Multiple Payment Methods
- Notifications
- Menu Management
- Sales Reports
- System Performance

Non-Functional Requirements:

- Performance Requirements
- Security Requirements
- Usability Requirements
- Reliability Requirements
- Maintainability Requirements
- Portability Requirements

Complexity: Medium

2. Architecture

Style: Monolithic

Pattern: Model-View-Controller (MVC)

Justification:

Given the medium complexity of the system, a monolithic architecture is more appropriate as it simplifies the management and communication between components. The MVC design pattern will allow for separation of concerns: Model (data), View (presentation), Controller (handle user input). This structure promotes maintainability, improves reusability of code, and facilitates scalability in the future if needed.

3. Tech Stack

Frontend: React.js

Backend: Node.js with Express.js

Database: PostgreSQL

Tools:

- Webpack
- Jest

Justification:

1) React.js is a popular JavaScript library for building user interfaces, providing fast render times and strong community support. It's also compatible with the MVC design pattern. 2) Node.js with

Express.js serves as an efficient backend solution, offering non-blocking I/O operations and a robust framework for handling HTTP requests and responses. 3) PostgreSQL is a powerful, open-source relational database management system that can handle complex queries, offers high performance, and provides strong security features. It's also compatible with Node.js. 4) Webpack is an excellent module bundler for managing assets in the frontend, while Jest is a popular testing framework for both the frontend and backend.

4. Project Structure

```
...
/myrestaurantapp
  frontend
    public
      index.html
    src
      components
        Auth
        Login
        Register
        Navbar
        Restaurant
        Browser
        Menu
        OrderTracking
        SalesReport
        Notifications
        Payments
        Orders
        App.js
        index.js
      packages.json
      webpack.config.js
  backend
    controllers
      authController.js
      restaurantController.js
      orderController.js
      paymentController.js
      notificationController.js
    models
      authModel.js
      restaurantModel.js
      orderModel.js
      paymentModel.js
      notificationModel.js
    routes
      authRoutes.js
      restaurantRoutes.js
      orderRoutes.js
      paymentRoutes.js
      notificationRoutes.js
    server.js
    package.json
  postgresql
    schema.sql
```

